



PCF lab Certificate Rotation Estimate

- **Director:** 192.168.2.2
- **Generated:** 2026-06-10 11:12:17 UTC
- **Horizon:** all certificates (no expiry limit)
- **Topology source:** maestro tp
- **Update rules:** bosh manifest (per-IG max_in_flight/canaries)

Summary

Metric	Value
Expiring certs (all (no limit))	112 (✗ 0 · ⚠ 0)
— leaf / CA	105 / 7
Live VMs (DB nodes)	7 (2)
One foundation-wide Apply	1h 00m – 1h 50m
Estimated rotation time	3h 00m – 5h 30m

Foundation inventory

Deployment	Type	VMs	Est. cert-rotation time
cf-dbe1a7580979a87638e7	cf	7	3h 00m – 5h 30m

2 instance group(s) have serial:false — they may update in parallel, so real time can come in under the (conservative serial) estimate.

CA rotations (3-phase)

Certificate	Expires
<code>.properties.nats_client_ca.3506e33356bef86af963</code>	2030-05-29T03:06:57Z
<code>.properties.root_ca.3506e33356bef86af963</code>	2030-05-29T03:06:57Z
<code>/p-bosh/cf-dbe1a7580979a87638e7/diego-instance-identity-intermediate-ca-2-7</code>	2028-06-09T04:19:47Z
<code>/opsmgr/bosh_dns/tls_ca</code>	2030-05-29T16:32:50Z
<code>/cf/diego-instance-identity-root-ca-2-6</code>	2029-05-29T16:32:45Z
<code>/services/tls_ca</code>	2031-05-29T03:24:34Z
<code>opsman-root-ca:3506e333</code>	2030-05-29T03:06:57Z

Leaf certificates (1-phase)

Deployment	Leaf certs	Apply
<code>p-bosh (not VM-counted)</code>	11	folded into shared apply #2
<code>cf-dbe1a7580979a87638e7</code>	94	folded into shared apply #2

Estimate breakdown

Campaign	Applies	Time (low – high)
Leaf certs	folded into the shared apply #2	included
CA certs (full batch)	3× foundation-wide on all tiles — all CAs + leaves	3h 00m – 5h 30m
Total		3h 00m – 5h 30m

Model & assumptions

- 20m overhead per Apply Changes; stateless 4–10m/VM, DB 10–20m/node.
- A FOUNDATION CA = 3 foundation-wide applies; a services TLS CA = 2 foundation + 1 deployment applies; a deployment CA = 3 on its deployment.

- Estimate is **Apply Changes compute time only** — it excludes change-window / approval gaps between phases, which often dominate the wall-clock for CA rotations.